82. Sun's halo is produced by the refraction of light in:

(a) Water vapour in Stratus clouds.
(b) Ice crystals in Cirro-Cumulus clouds.
(c) Ice crystals in Cirrus clouds.
(d) Dust particles in Stratus clouds.

Ans. (c) : 'Sun Halo', a circle of light with 22º radius, is Produced when sun light is refracted in hexagonal suspended ice crystals and super cool water droplets formed in the cirrus clouds.

CHEMISTRY

83. Which one of the following statements is not correct?

(a) The boiling point of an aqueous solution is higher than that of pure water.
(b) Addition of solutes to a solution causes an increase in its water potential.
(c) The vapour pressure of the water in a solution is lower than that of pure water.
(d) When a solution is separated from water by a semi permeable membrane, water movement can be prevented by applying pressure to the solution.

Ans. (b) : On adding a solute or other soluble substance to water, the boiling point increases because the solute molecule present on the surface reduces the rate of evaporation of water. Due to this, the vapour pressure of water in the solution is lower than that of pure water. The entry of solvent into the solution can be prevented by osmotic pressure, that is, external pressure on the solution. Thus the statements a, c, d are true and (b) false.

84. Assertion (A) : Synthetic detergents can lather well in hard water.

Reason (R) : Synthetic detergents form soluble calcium and magnesium salts with hard water.

(a) Both A and R are individually true and R is the correct explanation of A.
(b) Both A and R are individually true but R is NOT the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

Ans. (a) : Synthetic detergents are used in the case of hard water because the calcium and magnesium salts of detergents are soluble in water. Detergents are more soluble than soaps and hence form more lather than soaps. Therefore both Assertion and Reason are correct and Reason is the correct explanation for Assertion.

85. Match List I with List II and select the correct answer using the code given below the lists:

List I (Oxidation No.)	List II (The element)
A. 2	1. Oxidation number of Mn in MnO_2.
B. 3	2. Oxidation number of S in $H_2S_2O_7$.
C. 4	3. Oxidation number of Ca in CaO_2.
D. 6	4. Oxidation number of AI in $NaAlH_4$.

	A	B	C	D
(a)	3	4	1	2
(b)	4	3	1	2
(c)	3	4	2	1
(d)	4	3	2	1

Ans. (a) : The oxidation number of an element is the number which tells the number of charges on that atom in a molecule or ion, if all the remaining atoms from that molecule or ion are dissociated as potential ions. Oxidation number of Mn in MnO_2 is 4, oxidation number of S in $H_2S_2O_7$ is 6, Oxidation number of Ca in CaO_2 is 2 and Oxidation number of AL in $NaALH_4$ is 3.

BIOLOGY

86. With reference to the latest developments in stem cell research, consider the following statements:

1. **The only source of human stem cells are the embryos at blastocyst state.**
2. **The stem cells can be derived without causing destruction to blastocysts.**
3. **Then stem cells can regenerate themselves in vitro virtually forever.**
4. **Indian research centres also created a few cell lines which can be developed into many types of tissues.**

Which of these statements are correct?

(a) 1, 2, 3 and 4 (b) 1, 2 and 3
(c) 1, 2 and 4 (d) 3 and 4

Ans. (a) : Embryos at the cystic stage are the only source of human stem cells. Stem cells can be obtained without destroying the cysts. Stem cells can literally survive on their own for eternity. Indian research centers have also created some cell lines that can be grown into many types of tissues. Hence all the four statements are correct.

87. Which one of the following statements is not correct?

(a) All echinoderms are viviparous.
(b) Roundworm has no circulatory system.
(c) In bony fishes, swim bladder is usually present.
(d) In cartilaginous fishes, fertilization is internal.

Ans. (a) : The main characteristics of the organisms of Echinodermata Union are as follows: (i) All organisms of this union are marine. (ii) Water convection system is found in it. (iii) The brain does not develop in the nervous system. (iv) Special ability to reproduce. Examples - star fish, sea urchin, sea cucumber, feathers, britrale star, (v) Most echinoderms are oviparous.

88. Match List I with List II and select the correct answer using the codes given below the lists:

List I (Scientists)	List II (Achievement)
A. Arber and Smith	1. Developed transgenic plants with Agro bacterium T-DNA.
B. Feldman	2. Discovered endonucleases.
C. Mullis	3. Discovered reverse transcriptase.
D. Temin and Baltimore	4. Discovered polymerase chain reaction.

	A	B	C	D
(a)	2	1	4	3
(b)	1	2	4	3
(c)	2	1	3	4
(d)	1	2	3	4

Ans. (a) : Arber and Smith explored the restriction enzymes. Feldman developed parasitic plants containing Agro bacterium (T - DNA). Mullis detected the polymerase chain reaction (PCR), and Tamin and Baltimore detected reverse transcriptase.

89. Which one of the following statements is correct?

(a) Prions are the smallest free-living cells.
(b) The cell wall of Mycoplasmas is made up of amino sugars.
(c) Viroids consist of a single stranded RNA molecule.
(d) Rickettsias lack cell wall.

Ans. (c) : Viroid is a small infective segment of nucleic acid, usually RNA. Prion is an infectious pathogenic agent devoid of nucleic acid and composed mainly of proteins. Mycoplasmas are smallest free living cell and they lack a cell wall while Rickettsiase possess a cell wall.

Direction : The following 10 (ten) items consist of two statements, one labelled as the 'Assertion (A) and the other as 'Reason (R). You are to examine these two statements carefully and select the answers to these items using the code given below:

(a) Both A and R are individually true and R is the correct explanation of A.
(b) Both A and R are individually true but R is NOT the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

90. Assertion (A) : Drinking of whisky increases the frequency of urination.

Reason (R) : Alcohol intake speeds up the secretion of vasopressin in the body.

(a) Both A and R are individually true and R is the correct explanation of A.
(b) Both A and R are individually true but R is NOT the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

Ans. (c) : Alcohol acts as a diuretic. It actually promotes urine production by inhibiting the release of antidiuretic hormone, or vasopressin from the pituitary gland. In turn, reduced levels of antidiuretic hormone, prevent the kidneys from reabsorbing water and thereby increase urine production.

91. Assertion (A) : Human diet should compulsorily contain Glycine, Serine and Tyrosine.

Reason (R) : Essential amino acids cannot be synthesized in the human body.

(a) Both A and R are individually true and R is the correct explanation of A.
(b) Both A and R are individually true but R is NOT the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

Ans. (d) : Glycine, Serine and Tyrosine are non-essential Amino Acids. That means human body can produce them even if it is not provided from food. Hence (A) is false. But R is true as essential Amino Acids cannot be made by the body. Hence they must come from food.

92. Ticks and mites are actually

(a) Arachnids (b) Crustaceans
(c) Insects (d) Myriapods